From Useful to Actionable Information: A Task-Relevant Information Hierarchy for Ultrasonic Inspection of Impacted Composites

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Abstract

Ultrasonic inspection produces spatially rich internal observations, but predictive usefulness alone does not determine which measurements should be acquired. We introduce a task-relevant information hierarchy that distinguishes useful, observable, and actionable information. Usefulness asks whether an observation improves a downstream task; observability asks whether its conditional value can be estimated from the legally available inspection state; and actionability asks whether estimated value improves a bounded sensing decision. The hierarchy is evaluated on 276 specimens from six experimental domains using strict nested leave-one-domain-out validation and exact native-raster acquisition cost. A selected spatial ultrasonic field reduced equal-domain compression-after-impact prediction error by 0.06072 (32.1%) relative to its matched scalar representation, while a registered sparse condition retained 89.9% of the full-field gain. Retrospective oracle designs also differed between mechanical and reconstruction objectives. These useful signals did not transfer automatically to acquisition. The retrospective teacher's best action changed for 70.4% of specimens as measurements accumulated, yet static value ranking was unsupported, measured-content controls were adverse, and shuffled content remained stronger than real content for the tested dynamic scorer. Feedback was adverse, and the frozen learned policy did not outperform the strongest deployable baseline. The results establish a practical evidence hierarchy for separating information opportunity from observable measurement value and end-to-end sensing benefit.

Keywords: adaptive ultrasonic inspection, compression after impact, value of information, task-driven sensing, domain generalization

1. Introduction

Ultrasonic inspection produces spatially rich internal observations of impacted composites, but an engineering decision rarely needs every observable detail. The practical question is which measurements reduce uncertainty about a specified performance quantity enough to justify their acquisition. For carbon-fiber-reinforced polymer (CFRP) structures, this distinction is important because low-velocity impact may leave limited surface evidence while producing distributed internal damage that affects compression-after-impact (CAI) performance. Surface-to-damage and surface-to-CAI prediction studies show that accessible geometry carries useful information [1, 2], whereas image-resolved C-scan studies show that internal morphology provides a richer representation [3, 4]. Neither result alone determines where a future inspection should measure.

Sparse reconstruction and adaptive sensing offer possible reductions in digital measurement count. Existing work has demonstrated sparse ultrasonic reconstruction [5], adaptive path planning for defect localization [6], and autonomous task-directed inspection [7]. More generally, task-driven experimental design selects measurements according to downstream features rather than generic image fidelity [8]. These developments motivate adaptive acquisition, but they leave a crucial evidential gap: a measurement can contain task-relevant information without its value being inferable from the partial state available before acquisition, and an estimated value can still fail to improve a cost-constrained policy.

This paper formalizes that gap as a three-layer task-relevant information hierarchy. *Useful* information improves a specified downstream predictor under a held-out-domain protocol. *Observable* information has conditional value that can be recovered from the legal inspection state without future outcomes or unrevealed measurements. *Actionable* information changes a bounded acquisition decision enough to improve end-to-end utility. The layers are deliberately non-equivalent: useful does not imply observable, and observable does not imply actionable. This framing adapts value-of-information reasoning [9, 10] to sequential ultrasonic measurement while keeping the downstream predictor, acquisition state, action set, and cost definition explicit.

We examine three research questions. RQ1 asks whether spatial and sparse ultrasonic representations contain CAI-relevant information beyond matched lower-dimensional observations. RQ2 asks whether conditional measurement value is observable from legal partial-inspection states across unseen experimental domains. RQ3 asks whether the resulting valuation and bounded planning improve cost-constrained acquisition relative to deployable references. The empirical case study uses 276 paired specimens from six CFRP experimental domains, strict nested leave-one-domain-out evaluation, specimen-first inference, and exact native-raster location cost. Retrospective teachers and oracles are used only to measure opportunity or diagnose a component; they are never presented as deployable policies.

The contribution is fourfold. First, the hierarchy supplies a precise vocabulary and operational test for separating task usefulness, conditional observability, and decision actionability. Second, matched scalar, surface, sparse-field, full-field, and cross-objective oracle contrasts quantify where useful information exists and where its interpretation must remain bounded. Third, static, dynamic, position-only, reconstruction, and shuffled-content controls test whether partial-state valuation is attributable to measured specimen content. Fourth, retrospective component substitutions, bounded set planning, feedback, and a frozen end-to-end comparison show how diagnostic headroom can coexist with a negative deployment-facing result. The outcome is not a claim of a superior acquisition model; it is an evidence-centered framework for deciding which information claims the current data and validation protocol support.

2. Related Research and Problem Formulation

2.1. Ultrasonic information for post-impact performance assessment

Low-velocity impact can leave limited visible evidence while producing a spatially distributed internal damage state that affects compression-after-impact (CAI) performance. Surface profiles have consequently been studied as accessible predictors of internal impact damage and residual CAI strength [1, 2]. These mappings establish that surface observations can be useful within a defined cohort, but they do not make the surface and internal fields informationally equivalent. Image-resolved analyses of impacted laminates likewise motivate retaining spatial morphology rather than reducing damage to a single projected extent [3].

The paired public releases used here provide specimen metadata, surface profiles, ultrasonic C-scans, and CAI measurements [11, 12]. Mack et al. subsequently showed that full C-scan images can support direct prediction of impact energy and CAI strength with a convolutional model [4]. That study is the closest application-level precedent: it establishes full-image predictive feasibility, so the present contribution is not the use of C-scans or image features for CAI regression. The unresolved engineering-information question is narrower and different. It asks which portions of the measured field carry task value, whether the value of an unacquired portion can be inferred from a legal partial inspection state, and whether that estimate improves the next measurement decision across held-out experimental domains.

This distinction also separates a predictive input from an acquisition policy. A full-field model may quantify the information available after the inspection has ended, but it does not tell an inspector which location to measure next or whether the value assigned to that location can be known before measurement. Those propositions require different information sets and different controls.

2.2. Sparse and adaptive ultrasonic inspection

Sparse ultrasonic sampling has been used to recover impact-delamination patterns from incomplete measurements [5]. Recent adaptive sampling of composite Lamb wavefields combines a damage-guided sampling pattern with a spatial–temporal masked autoencoder, with full-wavefield reconstruction as the learning endpoint [6]. Autonomous robotic ultrasonic inspection has also selected successive physical locations using Bayesian optimization over a damage-probability or novelty field [7]. Adaptive and sequential ultrasonic inspection are therefore established; neither is a priority claim of this work.

The acquisition objective matters. Reconstruction-oriented sampling rewards a measurement when it improves the recovered signal or image. Damage-localization sampling rewards evidence that resolves the presence or position of a defect. The present CAI task instead rewards measurements that reduce residual-capacity prediction error. These objectives can select the same locations, but this agreement cannot be assumed. A visually or numerically accurate reconstruction may preserve features irrelevant to CAI, whereas a mechanically useful measurement need not minimize global image error. We therefore compare mechanics- and reconstruction-conditioned utilities under a common acquisition and cost contract.

Sparse fractions in this paper refer to locations on a registered digital raster. They are not converted into scanner time. Physical travel, coupling, settling, pulse repetition, and path-planning overhead are absent from the released data, so a native-raster measurement reduction is not evidence of an equal inspection-time reduction.

2.3. Task-driven sensing and value of information

Task-driven experimental design explicitly chooses measurements for a downstream objective rather than for signal reconstruction alone. For example, TADRED selects a compact global channel design from dense multi-channel images while optimizing a user-specified task [8]. In ultrasonic structural health monitoring, value-of-information criteria have been used to trade diagnostic information against the number and placement cost of guided-wave sensors [10]. These works establish both task-driven imaging and ultrasonic value-of-information design as prior art.

Sequential decision making adds a further dependency: the value of a future observation depends on what is already known. In a partially observable infrastructure-management setting, component inspection, monitoring, and system-level scheduling are distinct decisions because beliefs and future opportunities evolve after observations [9]. This separation is directly relevant to partial C-scan inspection. A retrospective calculation may show that a location would have reduced CAI error, while a deployable score may fail to identify that location from the current state; even an accurate one-step score may fail when measurements must be selected as a cost-constrained set.

Accordingly, this paper does not present task-driven acquisition itself as the novelty. Its contribution is an evidence hierarchy that separates three claims often collapsed in engineering machine learning: whether information is useful to a task, whether its conditional value is observable before acquisition, and whether the resulting estimate is actionable in a bounded sensing decision.

2.4. Problem formulation and research questions

Let specimen i belong to experimental domain $d(i)$. Its response y_i is the damaged-to-intact CAI-strength ratio. Let z_i denote legal pre-inspection context, S_i the complete native-raster ultrasonic field, and $\mathcal{I}_{i,t}$ the information legally available after acquisition step t. The latter contains z_i, acquired coordinates and values, action history, and exact cost, but excludes unrevealed values in S_i and the outcome y_i. A candidate measurement $X \in \mathcal{A}(\mathcal{I}_{i,t})$ is legal only if it refines the current hierarchical measurement state without exceeding the budget.

For a fixed downstream predictor f and task loss ℓ, define the risk of a state as

$$\mathcal{R}_f(\mathcal{I}_{i,t}) = \ell(y_i, f(\mathcal{I}_{i,t})) \,. \tag{1}$$

The realized task value of candidate X is

$$U_f(X \mid \mathcal{I}_{i,t}) = \mathcal{R}_f(\mathcal{I}_{i,t}) - \mathcal{R}_f(\mathcal{I}_{i,t} \cup X). \tag{2}$$

A positive value means that revealing X reduces error for this predictor at this state. Because both terms depend on f, Eq. (2) defines *downstream-predictor-conditioned task value*; it is not an intrinsic or universal mechanical property of a location. During retrospective analysis, U_f may use the true response and counterfactual revealed measurements. Such values are teacher or oracle evidence and are unavailable to a deployable policy.

The study addresses three research questions:

RQ1 — Useful: What ultrasonic information is useful for CAI assessment, and does the optimal acquisition objective depend on the downstream task?

RQ2 — Observable: Can the value of a future measurement be inferred from the information legally available at the current inspection state?

RQ3 — Actionable: Can conditionally valued information be converted into a better cost-constrained sensing decision?

These questions are ordered but not logically equivalent. Evidence for RQ1 provides a target for RQ2; it does not guarantee that the target can be estimated. Evidence for RQ2 provides a score for RQ3; it does not guarantee that greedy, planned, or feedback-driven selections improve the final cost–error trajectory.

3. Task-Relevant Information Hierarchy and Operational Framework

3.1. Framework overview

The Task-Relevant Information Hierarchy in Fig. 1 treats Useful, Observable, and Actionable as separate empirical layers. At the first layer, a measurement is useful if access to it improves a defined engineering task under a fixed predictor and validation protocol. At the second, its conditional value is observable if a score computed only from legal current information agrees with a strict out-of-fold retrospective target. At the third, the score is actionable if using it in a cost-constrained policy improves the complete decision trajectory relative to deployable controls.

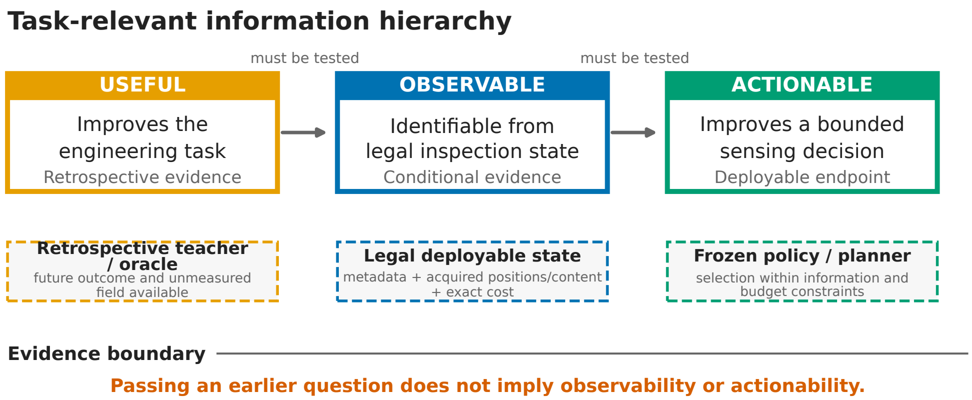

Figure 1: Task-relevant information hierarchy. Useful information improves the engineering task, observable information can be inferred from legally available inspection state, and actionable information improves a bounded sensing decision. Retrospective teacher/oracle evidence is separated from the deployable state and policy; passing one question does not imply the next.

The hierarchy is deliberately fail-closed. A result at one layer authorizes a question at the next layer, not a success claim. This prevents a retrospective oracle from being presented as a policy, a high rank correlation from being presented as improved acquisition, or lower reconstruction error from being presented as higher structural-performance value.

3.2. Downstream-predictor-conditioned task usefulness

Usefulness is assessed by changing the information available to the downstream task while holding the cohort, held-out domains, response, and validation logic fixed. The first comparison asks whether a spatial internal field adds CAI-relevant information beyond a matched scalar representation. The second asks how much of the registered full-field gain remains under a fixed sparse sampling and interpolation protocol. These are predictive information comparisons; neither identifies a deployable adaptive acquisition rule.

For candidate designs, the corresponding retrospective oracle chooses

$$X^*_{f,i,t} = \arg \max_{X \in \mathcal{A}(\mathcal{I}_{i,t})} U_f(X \mid \mathcal{I}_{i,t}) \quad \text{subject to} \quad c(X \mid \mathcal{I}_{i,t}) \leq b_t, \tag{3}$$

where c is exact added native-raster cost and b_t is the remaining budget. Because evaluation of Eq. (3) requires counterfactual measurements and y_i, it measures headroom rather than deployability. Applying the same acquisition contract to CAI error and image-reconstruction error tests whether utility is task-specific. Repeating the value calculation with different downstream predictors tests whether the map is stable to the definition of f.

3.3. Conditional observability

Observability concerns an estimate, not the retrospective value itself. Let g_θ be a scorer with inputs restricted to the legal state and a candidate descriptor. Its prediction is

$$\widehat{U}_{i,t,X} = g_\theta(\mathcal{I}_{i,t}, X, c(X \mid \mathcal{I}_{i,t}))\,. \tag{4}$$

Training labels are generated strictly out of fold: for each outer target domain, teacher predictors and scorer selection use source domains only. The held-out domain is queried after all fitting and selection are fixed. Agreement with the teacher is examined through within-specimen candidate ranking, best-action and top-k agreement, and exact-budget set regret.

Conditional observability has two potential failure modes. First, a static pre-acquisition representation may contain insufficient specimen-specific information to rank future measurements. Second, the target itself may evolve as observations accumulate. We therefore measure teacher-value turnover across states before asking whether a dynamic scorer tracks it. Matched positions-only, reconstruction, static, and shuffled-content controls distinguish access to acquisition geometry from evidence attributable to newly measured specimen content. These controls bound the claim: improvement relative to an empty state does not establish content-specific observability if geometry or shuffled content performs as well or better.

3.4. Decision actionability

Actionability evaluates the policy produced by a score. At state $\mathcal{I}_{i,t}$, a deployable selector chooses a legal candidate using current information and exact incremental cost. A one-step value-per-cost rule is

$$\pi(\mathcal{I}_{i,t}) = \arg\max_{X\in\mathcal{A}(\mathcal{I}_{i,t})} \frac{\widehat{U}_{i,t,X}}{c(X \mid \mathcal{I}_{i,t})}. \tag{5}$$

The registered testbed also evaluates direct cost-aware scores and bounded lookahead, but all choices obey the same reveal and budget contract.

Policy quality is measured over the complete error–budget curve rather than at one convenient endpoint. For checkpoints $q_1 < \cdots < q_K$ with CAI errors $e_i(q_k)$, the per-specimen area under the error–budget curve is computed by trapezoidal integration,

$$\mathrm{AUEBC}_i = \sum_{k=1}^{K-1} \frac{e_i(q_k) + e_i(q_{k+1})}{2} (q_{k+1} - q_k), \tag{6}$$

with lower values preferred. Component substitutions compare learned and retrospective valuation or planning under a bounded diagnostic. They locate decision headroom but remain non-deployable when true future values enter. Feedback is tested separately because re-scoring after each reveal can help only if the state update carries usable specimen-specific information and the scorer responds to it appropriately.

3.5. Operationalization with causal partial-state measurement valuation

MAVIS is used here as a frozen operational testbed for state-conditioned valuation and actionability, not as a demonstrated superior adaptive scanner. Its authority layer issues two non-overlapping views. The deployable policy context contains specimen metadata and surface/context features but contains no complete scan or CAI outcome. A separate source-teacher view exposes the full scan and outcome only to source-domain teacher construction and retrospective evaluation. The held-out evaluation view is not accepted by policy or state encoder interfaces.

Causal reveal begins with a geometry-neutral uniform scout. The native raster is partitioned into 64 cells; each cell has a three-level refinement state. Only coordinates implied by the initial scout and subsequent legal actions are materialized. Revealed coordinates and RGB values, measurement levels, action history, native shape, and exact acquired count form an immutable inspection state. A state bank records these histories at the registered checkpoints.

The current-state encoder aggregates variable-length measured tokens with a permutation-invariant DeepSets model and fuses them with context and exact-cost features. The mechanics head predicts CAI from the same causally available state. For source domains, a strict-OOF teacher evaluates each legal candidate using

$$u_{i,t,X} = |y_i - \widehat{y}_i(\mathcal{I}_{i,t})| - |y_i - \widehat{y}_i(\mathcal{I}_{i,t} \cup X)|\,. \tag{7}$$

The dynamic scorer receives the encoded current state and candidate geometry, level transition, exact added cost, and remaining cost. It predicts both a raw selection score and the teacher value. Rollout applies the selected legal action, reveals only its registered coordinates, updates the state, and either re-scores with feedback or retains the frozen initial ordering for the no-feedback control.

This implementation makes the information boundary executable. Future scan content and the true CAI outcome are necessary for teacher labels and oracle diagnostics, but they cannot enter the deployable scorer or rollout. The observability and actionability claims are consequently evaluated against the same causal state that would be available at the corresponding digital inspection step.

4. Multi-Domain CFRP Case Study and Experimental Design

4.1. Specimens and ultrasonic/CAI measurements

The case study uses 276 CAI-complete carbon-fiber-reinforced polymer specimens from six released experimental datasets [11, 12]. The domains contain 45, 49, 43, 59, 42, and 38 specimens, respectively. They differ in laminate configuration and experimental conditions and are retained as indivisible validation units. The response is the damaged-to-intact CAI-strength ratio. Available modalities are specimen metadata, surface observables, an ultrasonic RGB C-scan of the internal damage field, and the CAI outcome.

The registered scan manifest binds every specimen identifier to its domain, native raster shape, pixel count, and source-content hashes. Native rasters are not resized for acquisition-cost accounting. This matters because scan sizes vary across the cohort: a nominal percentage alone does not specify the number of newly observed locations. Table 1 summarizes the cohort and evaluation contract.

4.2. Information representations and candidate measurements

The usefulness experiments compare three information resolutions. The scalar reference combines metadata, surface observations, and scalar summaries of the internal scan. The matched spatial candidate uses the same registered predictor family but retains the selected full C-scan field representation. A separate internal-only field estimator is reported only as a sensitivity path; it is not substituted for the matched confirmatory comparison. The sparse retrospective condition reveals a registered 25% sampling pattern on a

Table 1: Multi-domain CFRP case study and evaluation protocol.

Item	Protocol value
Physical cohort	276 CAI-complete physical specimens
Held-out domains	6 experimental domains
Domain specimen counts	74t7kcdgkr: 45; cgtnjyggtm: 49; w68dtmpfyf: 43; xcmzfsbd9t: 59; yfxyg8jm46: 42; ykhs7s2dck: 38
Available modalities	specimen metadata; surface observables; ultrasonic RGB C-scan internal field; CAI-ratio outcome
Validation protocol	strict nested leave-one-domain-out (LODO); outer-domain outcomes excluded from training and selection
Registered checkpoints	3.125%, 6.25%, 9.375%, 12.5%, 18.75%, 25%
Acquisition cost	exact unique newly observed native-raster locations divided by native count; no scanner-time equivalence
Teacher/oracle information	future outcomes, unmeasured fields, or counterfactual utilities; retrospective and non-deployable
Deployable information	specimen metadata, acquired positions, measured content, legal actions, and current exact cost
Statistical units	physical specimen first; held-out domain second; six domains receive equal weight
Bootstrap aggregation	specimens resampled within domain with synchronized contrasts; domain means then equally weighted
Computational rows	state-action and checkpoint rows are repeated computational records, not independent statistical samples

normalized raster and uses bilinear interpolation before applying the frozen field pathway.

For sequential acquisition, each native raster is divided into an 8×8 cell grid. A cell may move through three registered sampling levels, and an action refines exactly one cell by one level. Candidate descriptors contain the cell position, level transition, exact number of newly observed native locations, native count, and remaining exact budget. This representation permits comparison of uniform, random, reconstruction-conditioned, mechanics-conditioned, static, and state-conditioned choices without changing the legal action space.

Mechanics utility is the change in CAI prediction loss in Eq. (2). Reconstruction utility uses normalized RGB mean squared error over the same registered raster. These utilities are kept separate: an acquisition may be optimal for one and suboptimal for the other.

4.3. Whole-dataset generalization protocol

All headline results use leave-one-domain-out (LODO) evaluation. Each of the six experimental datasets serves once as the outer target domain. Model fitting, normalization, hyperparameter or checkpoint selection, teacher construction, policy choice, and fallback thresholds use only the other five source domains. The outer-domain outcomes are excluded until the frozen candidate is evaluated. When an inner selection loop is required, entire source domains remain intact. This nested structure avoids the optimistic error estimates that arise when model selection and assessment reuse observations [13] and aligns the validation unit with the intended unseen-experiment question rather than a specimen-random split [14].

The physical specimen is the primary statistical unit. Multiple candidate actions, random seeds, and checkpoints for one specimen are repeated computational records, not independent samples. Domain effects are first computed from paired specimen outcomes inside each held-out domain and then aggregated with equal weight across the six domains.

4.4. Privileged, deployable, and forbidden information

The information boundary is defined by when and where a variable is available. Metadata and surface observables are legal before ultrasonic acquisition. Acquired positions, measured RGB values, measurement levels, legal actions, and exact current cost are deployable state variables. Reconstructions from that legal state may be used as controls. Unacquired ultrasonic content and the true CAI outcome are forbidden to a deployable scorer or policy.

Teacher and oracle analyses deliberately use privileged information to define retrospective targets and headroom. Their outputs are never described as deployable. In particular, a true-value oracle may inspect counterfactual outcomes for all legal actions, and a joint near-oracle may compare reachable action sets. These rows answer whether an opportunity exists or where a bounded decision gap lies; they do not specify an implementable inspector.

4.5. Exact acquisition cost, metrics, and statistical inference

The registered checkpoints are 3.125%, 6.25%, 9.375%, 12.5%, 18.75%, and 25% of the native raster. Initial scouts are 1.5625% except in the one domain whose native geometry requires a 3.125% scout. At every transition, cost is the number of unique newly revealed native-raster coordinates. Effective budget is that count divided by the specimen's native pixel count. Previously observed coordinates have zero added cost, and an action that would exceed the current cap is illegal. No result is translated into physical scanner time.

For full-field and sparse prediction, the primary outcome is equal-domain mean absolute error in CAI ratio. Sequential CAI performance is summarized by AUEBC from Eq. (6); lower values are better. Reconstruction is measured by normalized RGB mean squared error. Observability metrics include Spearman rank agreement between predicted and strict-OOF teacher action values, best-action turnover or agreement, top-k overlap, and exact-budget set regret. Actionability contrasts use paired differences in specimen AUEBC.

Uncertainty intervals use synchronized bootstrap resampling of physical specimens within each held-out domain. The same resampled indices are applied to both methods in a contrast, domain means are recomputed, and the six domain means are equally weighted. The registered scalar-to-field family uses its predeclared familywise simultaneous interval. Other intervals retain the registered paired-bootstrap definitions. With only six domains, domain counts are always reported beside the aggregate effect rather than treated as a large-sample inferential substitute; this is consistent with the caution required for clustered data with few groups [15].

4.6. Validation matrix for RQ1–RQ3

RQ1 combines the matched scalar-to-spatial contrast, sparse-retention boundary, retrospective acquisition headroom, cross-objective oracle specificity, and downstream-predictor sensitivity. RQ2 combines static value observability, teacher-value evolution, matched positions/reconstruction controls, and dynamic/static/shuffled scoring controls. RQ3 combines retrospective component substitution, bounded set planning, feedback, and the frozen policy comparison against the strongest deployable baseline.

This matrix prevents evidence migration between layers. Oracle headroom cannot support a deployable policy claim; a dynamic score cannot support content-specific observability if its matched adverse controls remain stronger; and a planning diagnostic cannot support improved acquisition unless the frozen end-to-end policy also improves. Section 5 reports the three answers in this order.

5. Experimental Results and Discussion

5.1. RQ1: Task usefulness of spatial and sparse ultrasonic information

5.1.1. Registered scalar-to-spatial information contrast

The registered matched comparison supported a task-usefulness gain from spatial internal morphology. Equal-domain CAI-ratio MAE was 0.18920 for the scalar reference and 0.12849 for the selected spatial field, a reduction of 0.06072 (32.1%; familywise 95% CI 0.00664–0.15364). The field reduced error in five of six held-out domains. Relative to the metadata-and-surface reference, whose MAE was 0.18812, the same selected field reduced MAE by 0.05963 (31.7%; familywise 95% CI 0.00722–0.14842), again improving five domains. Thus the result is not an artifact of selecting an unusually weak scalar summary: both registered references lead to the same bounded conclusion that the spatial internal field contains CAI-relevant information absent from the lower-dimensional inputs.

A distinct internal-only field estimator produced a lower point MAE of 0.08964. It is retained as a sensitivity estimate, not substituted for the matched confirmatory path. The registered effect is therefore the 0.06072 reduction, not a cross-family difference constructed from the lowest available point estimate. Figure 2a shows the matched comparison.

5.1.2. Sparse retention and retrospective oracle headroom

The selected 25% bilinear sparse condition had MAE 0.13451. Relative to the surface reference, its reduction was 0.05361 (95% CI 0.00561–0.13544), with lower error in five of six domains. This condition retained 89.9% of the registered full-field gain. Retention was substantial but incomplete: sparse MAE remained 0.00602 above the selected full field (95% CI 0.00173–0.01083), and the sparse condition was worse in every held-out domain. The appropriate engineering conclusion is therefore that a registered sparse digital measurement retains most of the full-field task information, not that the sparse and full fields are equivalent or that physical scan time falls by the same percentage (Fig. 2b).

Retrospective oracle comparisons showed that the useful information was spatially heterogeneous. One-shot mechanical-oracle CAI AUEBC was 0.013458, compared with 0.017363 for uniform acquisition and 0.017188 for the one-shot reconstruction oracle. The corresponding improvements were 0.00391 (95% CI 0.00280–0.00502) and 0.00373 (95% CI 0.00284–0.00469), respectively, and both contrasts favored the mechanical oracle in all six domains. The

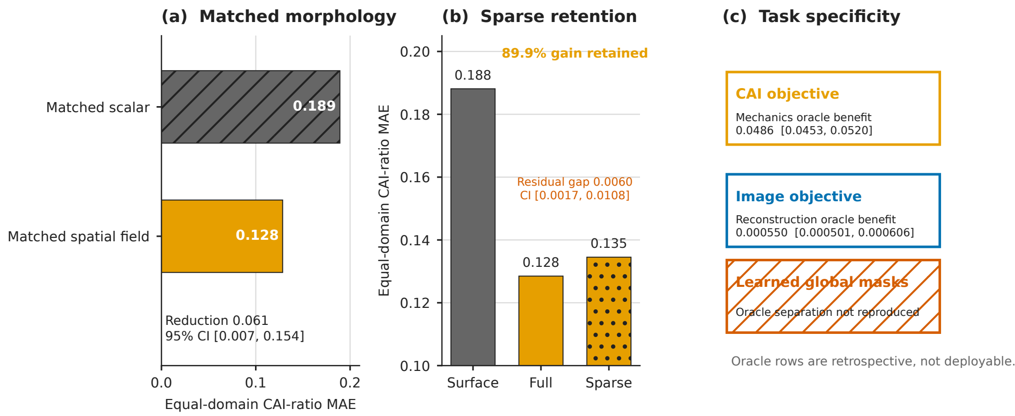

Figure 2: Usefulness evidence and boundaries. (a) The registered matched B-family spatial field reduces equal-domain CAI-ratio MAE relative to its scalar counterpart. (b) The selected sparse condition retains 89.9% of the registered full-field gain but remains worse than the full field. (c) Oracle acquisitions separate CAI and reconstruction objectives, whereas learned global masks do not reproduce that separation. Oracle and sparse-design rows are retrospective and do not establish scanner-time savings or deployability.

one-shot oracle retained 56.8% of the registered sequential-oracle headroom. These values establish retrospective specimen-specific opportunity, but the oracle uses unavailable counterfactual outcomes and is not a policy.

5.1.3. Mechanics versus reconstruction task specificity

Under the common exact-cost trajectory contract, oracle acquisition depended on the objective. The mechanics-optimal oracle improved CAI AUEBC relative to the reconstruction oracle by 0.04862 (95% CI 0.04527–0.05205). In the opposite direction, the reconstruction-optimal oracle improved normalized RGB image MSE by 5.503×10^{-4} (95% CI 5.006×10^{-4}–6.063×10^{-4}). The cross-objective contrasts therefore support task specificity at the retrospective oracle level (Fig. 2c).

The learned adverse control did not inherit this separation. Source-trained global mechanics and reconstruction masks produced a predeclared cross-objective support indicator of 0. The oracle result consequently shows that the objective changes the optimal design, but does not show that a learned global mechanics mask captures or deploys that distinction.

5.1.4. Predictor-conditioned task value

Action-value maps were partly stable across related predictors and unstable across a larger learner change. Ridge and Huber strict-OOF values had Spearman agreement 0.762 (95% CI 0.699–0.821), best-action agreement 0.675 (95% CI 0.622–0.727), and top-k Jaccard agreement 0.685 (95% CI 0.650–0.719). Ridge and shallow-MLP values instead had rank agreement 0.116 (95% CI 0.069–0.164), best-action agreement 0.233 (95% CI 0.182–0.283), and top-k agreement 0.213 (95% CI 0.192–0.235). The location ranking is therefore not a universal map of mechanical importance; it is conditional on the downstream predictor used to define improvement.

Answer to RQ1.. Yes. Spatial internal information improved held-out-domain CAI prediction, a registered sparse condition retained most but not all of that gain, and retrospective oracle designs differed between CAI and reconstruction tasks. However, learned global masks did not reproduce the oracle separation, and the value map changed with the downstream predictor. Task usefulness is therefore supported, but remains objective- and predictor-conditioned.

5.2. RQ2: Conditional observability during partial inspection

5.2.1. Static observability

The registered static scorer did not rank strict-OOF action values across held-out domains. Its equal-domain Spearman correlation was −0.0196 (95% CI −0.0591–0.0195), with a favorable direction in three of six domains (Fig. 3a). Exact-budget set regret was 0.08171 for the static scorer, compared with 0.07993 for the global mechanical score and 0.07978 for the random median. Increasing static scorer complexity was not authorized by these results: the registered representation had not first established a transferable specimen-specific target.

The confidence interval spans zero, so this result does not prove that value is information-theoretically unobservable. It shows that the tested pre-acquisition information and scorer did not recover the strict-OOF target under whole-domain holdout.

5.2.2. Conditional value evolution

The retrospective teacher target itself changed as information accumulated. Between the initial state and the 18.75% checkpoint, the best action changed for 70.4% of specimens. Spearman agreement with initial action values fell to 0.405, and top-five Jaccard overlap was 0.307. The descriptive dynamic-versus-initial opportunity at that checkpoint was 0.00531. These specimen-first diagnostics cover all 276 specimens across the six held-out domains and show that a single initial ranking omits material value evolution (Fig. 3b).

This is a property of the strict-OOF retrospective teacher, not evidence that the frozen real-state scorer observes the change. It motivates conditional scoring while making its validation harder: both the state representation and the scorer must track a moving, predictor-conditioned target.

5.2.3. Real-content, position, and reconstruction controls

Measured partial content did change the CAI prediction relative to the initial state. At 25%, real-state MAE changed by −0.000731 (95% CI −0.001184 to −0.000250), and the endpoint diagnostic recovered 26.6% (95% CI 10.6%–41.0%) of the static-to-independent-full-field error ratio. The latter denominator is the separate internal-only full-field sensitivity estimator, not the matched spatial estimator used for the RQ1 confirmatory result.

The matched controls reverse the favorable interpretation. Real-content MAE was 0.12463, compared with 0.10722 for positions only and 0.09043 for the reconstruction control. Real minus positions was 0.01740 (95% CI 0.00890–0.02582), and real minus reconstruction was 0.03419 (95% CI 0.02450–0.04384). Measured content was favorable in only one of six domains for each contrast (Fig. 3c). Thus real measurements add signal relative to an initial state, but the registered state representation does not establish specimen-specific content value beyond the acquisition geometry and reconstruction controls. The controls do not prove that geometry is universally sufficient; they delimit what this implementation identifies.

5.2.4. Narrow dynamic-valuation gain and shuffled adverse control

At the 18.75% endpoint, the dynamic real-state scorer reduced one-step value regret relative to the static scorer by 0.00126: dynamic minus static was −0.001260 (95% CI −0.002123 to −0.000444), favorable in five of six domains. This is the narrow positive observability result. It does not survive the content control. Dynamic real minus dynamic shuffled-content regret was 2.328×10^{-4} (95% CI 6.060×10^{-5}–4.180×10^{-4}), favorable for real content in only one domain. Shuffled content therefore remained better than the real-content scorer under the registered contrast.

Answer to RQ2.. Only partially. Strict-OOF conditional value evolved substantially and dynamic scoring narrowly improved on its static counterpart.

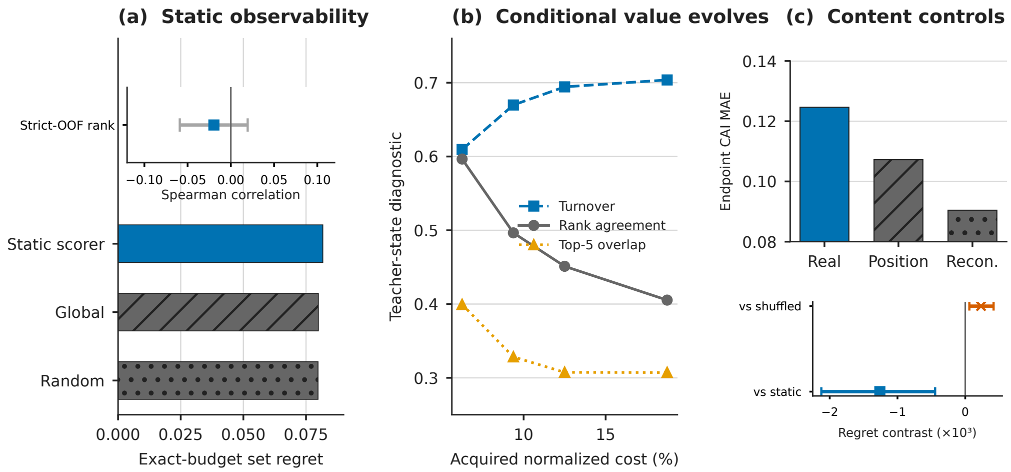

Figure 3: Observability evidence and adverse controls. (a) Strict-OOF static value ranking is unsupported and its exact-budget regret is comparable with global and random references. (b) Retrospective conditional values change with acquisition state. (c) Measured-content prediction is worse than matched-position and reconstruction controls; although conditional scoring narrowly improves on the static scorer, shuffled content remains better. Intervals are synchronized specimen-bootstrap intervals where shown.

Static ranking was unsupported, however, the real-content representation did not outperform matched position or reconstruction controls, and shuffled content remained the stronger dynamic control. Robust specimen-specific content-value observability is therefore not established by the current implementation.

5.3. RQ3: From valuation to cost-constrained sensing decisions

5.3.1. Valuation and planning attribution

Retrospective substitutions separated valuation from bounded planning effects (Fig. 4a). Replacing learned valuation with privileged retrospective values improved CAI AUEBC by 4.979×10^{-5} (95% CI 1.845×10^{-5}–8.105×10^{-5}). A bounded learned-planning substitution improved AUEBC by 3.164×10^{-6} (95% CI 1.411×10^{-7}–6.250×10^{-6}), whereas stronger planning with true values improved it by 1.117×10^{-4} (95% CI 9.898×10^{-5}–1.248×10^{-4}).

These contrasts show that valuation and planning can contribute distinct shortfalls. They do not prove that either retrospective substitution is deployable. A separate representation-bottleneck row was not interpreted because the matched observability controls in Section 5.2.3 did not establish content-specific state value.

5.3.2. Set-level planning gap

Within the predeclared two-action reachable pool, current greedy selection had regret 1.207×10^{-4} relative to a retrospective joint near-oracle set (95% CI 1.033×10^{-4}–1.386×10^{-4}). Beam search with width four reduced the point regret to 1.127×10^{-4}, but its interval remained positive (95% CI 9.485×10^{-5}–1.308×10^{-4}). The diagnostic therefore supports a bounded set-planning gap (Fig. 4b). It neither makes the joint near-oracle deployable nor establishes arbitrary-horizon planning regret.

5.3.3. Feedback stress test

Re-scoring after each reveal did not improve the frozen policy. With the registered sign convention of no-feedback error minus feedback error, the CAI AUEBC feedback benefit was -1.496×10^{-5} (95% CI -1.944×10^{-5} to -1.064×10^{-5}), and feedback improved two of six held-out domains (Fig. 4c). Negative values favor the no-feedback control, so feedback was adverse under this frozen implementation. Stratified diagnostics did not support larger conditional-value changes as the operative feedback mechanism. The result is implementation-bounded and does not imply that feedback is generally harmful.

5.3.4. Frozen cross-domain acquisition boundary

The frozen learned policy had equal-domain CAI AUEBC 0.125053, compared with 0.124992 for the strongest deployable baseline. Baseline minus learned-policy AUEBC was -6.114×10^{-5} (95% CI -8.461×10^{-5} to -3.777×10^{-5}), and the learned policy improved two of six held-out domains. Because lower AUEBC is preferred, this contrast does not support superiority of the learned policy.

The final boundary is stronger than an isolated null result. Retrospective measurement value and planning opportunity both exist, yet the current representation and scoring controls do not attribute that value to acquired specimen content, feedback is adverse, and end-to-end acquisition remains below the strongest deployable reference. Increasing model or planner complexity would not by itself resolve those evidential gaps.

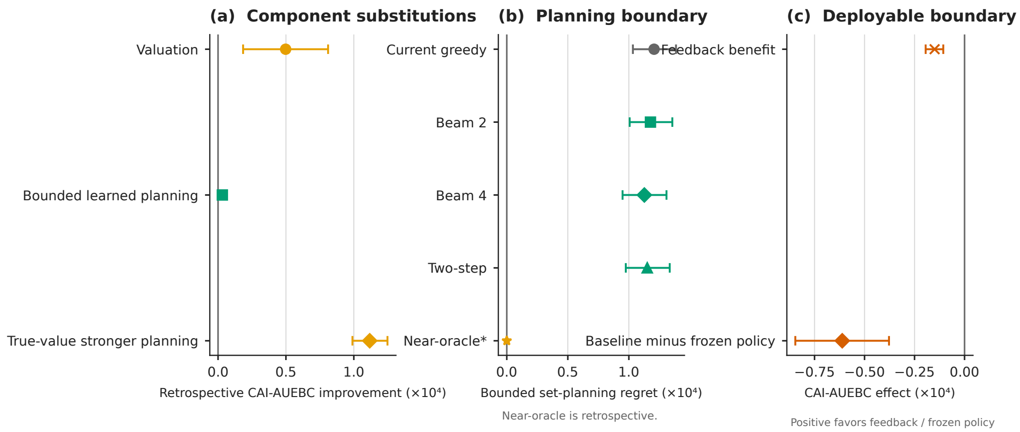

Figure 4: Actionability evidence and final boundary. (a) Retrospective component substitutions expose valuation and bounded planning gaps. (b) A positive two-action reachable-pool planning gap remains; the near-oracle is non-deployable. (c) Feedback is adverse and the frozen learned policy does not outperform the strongest deployable baseline. Effects use specimen-first, equal-domain aggregation and synchronized intervals where available.

Answer to RQ3.. No under the current learned implementation. Retrospective substitutions and a bounded near-oracle expose valuation and set-planning headroom, but feedback is adverse and the frozen policy does not outperform the strongest deployable baseline across held-out domains.

5.4. Integrated engineering-informatics interpretation

Table 2 assembles the evidence without treating the three layers as a single model-development sequence.

Table 2: Evidence summary for the task-relevant information hierarchy. Positive and negative directions are stated in the comparison text.

Layer	Question	Key comparison	Effect	95% CI	Domains	Evidence type	Conclusion
Useful	Does it improve the task?	matched scalar vs selected B-family spatial field	MAE reduction 0.0607 (32.1%)	[0.00664, 0.154]	5/6	registered confirmatory	Spatial internal morphology preserves CAI-relevant information absent from the matched scalar representation.
		surface vs sparse field; sparse vs full field	89.9% gain retained; full-field gap 0.00602	gap [0.00173, 0.0108]	5/6 vs surface; 0/6 vs full	registered retrospective sparse protocol	Sparse measurements retain most, but not all, registered full-field gain; no scanner-time claim is made.
		mechanical oracle vs uniform and reconstruction oracles	CAI-AUEBC improvements 0.00391 and 0.00373	[0.00280, 0.00502]; [0.00284, 0.00469]	6/6 for both	retrospective one-shot oracle	Specimen-specific mechanical acquisition headroom exists, but the oracle is non-deployable.
		mechanics and reconstruction task-specific oracles; learned global masks	CAI 0.0486; image error 5.503e-04; learned indicator 0	[0.0453, 0.0520]; [5.006e-04, 6.063e-04]	6-domain contrasts; learned separation unsupported	retrospective cross-objective oracle + learned adverse control	Oracle task specificity holds; learned global masks do not reproduce the separation.
		Ridge-Huber vs Ridge-MLP action-value rankings	Spearman rho 0.762 vs 0.116	[0.699, 0.821]; [0.0690, 0.164]	6 held-out domains	strict-OOF learner sensitivity	Measurement value is downstream-predictor-conditioned rather than an intrinsic location property.
Observable	Is value observable from legal state?	static legal-state scorer vs strict-OOF teacher values	Spearman rho -0.0196	[-0.0591, 0.0195]	3/6	strict-OOF static scorer	Static value observability is not supported.

Continued on next page

Table 2 (continued)

Layer	Question	Key comparison	Effect	95% CI	Domains	Evidence type	Conclusion
		conditional teacher values from initial to final tested checkpoint	turnover 0.704; rank 0.405; top-5 0.307	not estimated	276 specimens; 6 domains	strict-OOF retrospective teacher	True conditional measurement value evolves with acquisition state.
		measured content vs matched positions and reconstruction controls	real minus position MAE 0.0174; real minus reconstruction 0.0342	[0.00890, 0.0258]; [0.0245, 0.0438]	1/6 favors measured content for each control	matched representation adverse controls	Measured content is worse than positions and reconstruction controls; beyond-geometry value is not established.
		conditional real-state regret vs static and shuffled controls	vs static -0.00126; vs shuffled 2.328e-04	[-0.00212, -4.442e-04]; [6.060e-05, 4.180e-04]	5/6 vs static; 1/6 vs shuffled	frozen dynamic valuation controls	Conditional scoring narrowly beats static, but shuffled content remains better.
Actionable	Does it improve a bounded decision?	retrospective valuation and planning substitutions	valuation 4.979e-05; learned planning 3.164e-06; true-value planning 1.117e-04	[1.845e-05, 8.105e-05]; [1.411e-07, 6.250e-06]; [9.898e-05, 1.248e-04]	6-domain equal weighting	retrospective component substitution	Valuation and bounded planning gaps exist; a representation bottleneck is not established.
		current greedy vs retrospective joint near-oracle reachable pool	planning regret 1.207e-04	[1.033e-04, 1.386e-04]	6-domain equal weighting	retrospective bounded two-action planning	A bounded set-planning gap remains; the near-oracle is non-deployable.
		no-feedback minus feedback under the frozen policy	feedback benefit -1.496e-05	[-1.944e-05, -1.064e-05]	2/6	frozen feedback control	Feedback is adverse under the frozen cross-domain protocol.
		strongest deployable baseline minus frozen learned policy	CAI-AUEBC effect -6.114e-05	[-8.461e-05, -3.777e-05]	2/6	frozen deployable endpoint	The frozen learned policy did not outperform the strongest deployable baseline.

RQ1 establishes a real target for acquisition: spatial C-scan information improves CAI assessment, much of its registered gain survives a sparse digital protocol, and retrospective optimal designs change when utility changes from CAI error to image error. The predictor-sensitivity analysis sharpens the construct: task value belongs to a predictor, state, and objective, not to a coordinate alone.

RQ2 shows why that useful target is not automatically observable. The teacher ranking changes after measurements, so a static population map is incomplete. Yet state dependence alone is not sufficient evidence. The registered real-content representation loses to matched positions and reconstruction controls, and its narrow advantage over static scoring is contradicted by the shuffled-content control. In engineering-informatics terms, a model can respond to the structure of an inspection history without demonstrating that it has captured the specimen-specific content needed to value the next measurement.

RQ3 then separates observability from decision consequence. Privileged substitutions identify opportunities in valuation and bounded set planning, but neither is a deployable policy. The adverse feedback test and final AUEBC comparison show that those retrospective opportunities do not become a better frozen acquisition trajectory. The evidence therefore supports the hierarchy's central non-equivalence: Useful does not imply Observable, and partial Observable evidence does not imply Actionable improvement.

Four limitations bound this interpretation. First, the empirical evidence covers six related CFRP experimental domains from a paired public data program; external empirical generalization to other inspection systems, materials, or damage mechanisms is not established. Second, cost is exact native-raster location count, not travel time or industrial inspection cost. Third, the utility definition is specific to the chosen CAI predictors and cannot be read as causal attribution of failure mechanisms to selected image regions. Fourth, the action hierarchy, state representation, scorer, and bounded planners are fixed implementations. Their non-superiority does not prove that task-driven adaptive ultrasound is impossible; it shows which evidential conditions remain unmet before such a claim is warranted.

6. Conclusions

This study distinguished the existence of task-relevant ultrasonic information from the ability to observe and use its value during acquisition. For RQ1, Useful information was established: the registered spatial field improved held-out-domain CAI prediction over matched scalar and surface references, the sparse condition retained most but not all of the full-field gain, and retrospective optimal designs depended on the downstream objective and predictor. These findings identify a legitimate acquisition target without equating sparse digital coverage with scanner time or treating an oracle as a policy.

For RQ2, observability was only partial. Conditional teacher values changed substantially as measurements accumulated, and dynamic scoring narrowly reduced regret relative to static scoring. However, static ranking was unsupported, measured content was worse than matched positions and reconstruction controls, and shuffled content remained the stronger dynamic control. The tested state representation therefore did not establish transferable specimen-specific content value.

For RQ3, Actionable improvement was not established. Retrospective substitutions and bounded near-oracle planning exposed valuation and set-selection headroom, but feedback was adverse and the frozen learned policy did not outperform the strongest deployable baseline. The combined result is the hierarchy's central engineering-informatics conclusion: useful information may define what would be worth measuring in retrospect while remaining unobservable from a legal partial state and ineffective in an end-to-end acquisition policy. Progress should therefore be judged by matched information controls and frozen cross-domain policy outcomes, not by oracle opportunity or component-level improvements alone.

Data and code availability

The paired experimental datasets analyzed in this study are public sources cited in Section 4. An anonymized reproducibility package accompanies the submission and contains the paper-specific canonical metrics, source hashes, figure and table sources, supplemental machine-readable records, and deterministic source archive. The public code repository and archival identifier will be declared after review so that the review manuscript remains identity-neutral.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI Codex to assist with evidence auditing, deterministic analysis-code generation, LaTeX drafting, and language editing. The authors reviewed and validated all generated content against the original machine-readable artifacts and take full responsibility for the manuscript.

References

[1] S. Hasebe, R. Higuchi, T. Yokozeki, S.-i. Takeda, Internal low-velocity impact damage prediction in CFRP laminates using surface profiles and machine learning, Composites Part B: Engineering 237 (2022) 109844. doi:10.1016/j.compositesb.2022.109844.
URL https://doi.org/10.1016/j.compositesb.2022.109844

[2] S. Hasebe, R. Higuchi, T. Yokozeki, S.-i. Takeda, Prediction of compression after impact strength from surface profile of low-velocity impact damaged CFRP laminates using machine learning, Composites Part A: Applied Science and Manufacturing 189 (2025) 108560. doi:10.1016/j.compositesa.2024.108560.
URL https://doi.org/10.1016/j.compositesa.2024.108560

[3] D. J. Bull, S. M. Spearing, I. Sinclair, Image-enhanced modelling of residual compressive after impact strength in laminated composites, Composite Structures 192 (2018) 20–27. doi:10.1016/j.compstruct.2018.02.047.
URL https://doi.org/10.1016/j.compstruct.2018.02.047

[4] J. P. Mack, F. Mirza, Z.-H. Duan, S. Hasebe, R. Higuchi, T. Yokozeki, K. T. Tan, Deep learning for predicting impact energy and compression after impact strength of composite materials using C-scan images, Advanced Engineering Informatics 72 (2026) 104518. doi:10.1016/j.aei.2026.104518.
URL https://doi.org/10.1016/j.aei.2026.104518

[5] D. Nguyen, P. Davidson, K. DeMille, V. Ranatunga, Machine learning based segmentation of delamination patterns from sparse ultrasound data of barely visible impact damage in composites, Journal of Composite Materials 59 (2025) 321–330. doi:10.1177/00219983241292779.
URL https://doi.org/10.1177/00219983241292779

[6] D. Ji, W. Li, F. Gao, J. Hua, J. Lin, Adaptive sampling for efficient lamb wavefield reconstruction in composite laminates with spatial-temporal

masked autoencoder, Ultrasonics 162 (2026) 107972. doi:10.1016/j.ultras.2026.107972.
URL https://doi.org/10.1016/j.ultras.2026.107972

[7] R. Fuentes, P. Gardner, C. Mineo, T. J. Rogers, S. G. Pierce, K. Worden, N. Dervilis, E. J. Cross, Autonomous ultrasonic inspection using bayesian optimisation and robust outlier analysis, Mechanical Systems and Signal Processing 145 (2020) 106897. doi:10.1016/j.ymssp.2020.106897.
URL https://doi.org/10.1016/j.ymssp.2020.106897

[8] S. B. Blumberg, P. J. Slator, D. C. Alexander, Experimental design for multi-channel imaging via task-driven feature selection, International Conference on Learning Representations (2024). arXiv:2210.06891.
URL https://openreview.net/forum?id=MloaGA6WwX

[9] M. Memarzadeh, M. Pozzi, Value of information in sequential decision making: Component inspection, permanent monitoring and system-level scheduling, Reliability Engineering & System Safety 154 (2016) 137–151. doi:10.1016/j.ress.2016.05.014.
URL https://doi.org/10.1016/j.ress.2016.05.014

[10] S. Cantero-Chinchilla, J. Chiachío, M. Chiachío, D. Chronopoulos, A. Jones, Optimal sensor configuration for ultrasonic guided-wave inspection based on value of information, Mechanical Systems and Signal Processing 135 (2020) 106377. doi:10.1016/j.ymssp.2019.106377.
URL https://doi.org/10.1016/j.ymssp.2019.106377

[11] S. Hasebe, R. Higuchi, T. Yokozeki, S.-i. Takeda, Dataset for surface and internal damage after impact on CFRP laminates, Data in Brief 43 (2022) 108462. doi:10.1016/j.dib.2022.108462.
URL https://doi.org/10.1016/j.dib.2022.108462

[12] S. Hasebe, R. Higuchi, T. Yokozeki, S.-i. Takeda, Dataset of compression after impact testing on carbon fiber reinforced plastic laminates, Data in Brief 60 (2025) 111509. doi:10.1016/j.dib.2025.111509.
URL https://doi.org/10.1016/j.dib.2025.111509

[13] S. Varma, R. Simon, Bias in error estimation when using cross-validation for model selection, BMC Bioinformatics 7 (2006) 91. doi:10.1186/

1471-2105-7-91.
URL https://doi.org/10.1186/1471-2105-7-91

[14] I. Gulrajani, D. Lopez-Paz, In search of lost domain generalization, International Conference on Learning Representations (2021).
URL https://openreview.net/forum?id=lQdXeXDoWtI

[15] F. L. Huang, Using cluster bootstrapping to analyze nested data with a few clusters, Educational and Psychological Measurement 78 (2) (2018) 297–318. doi:10.1177/0013164416678980.
URL https://doi.org/10.1177/0013164416678980